

About me

Electronic engineer with experience in hardware design. Passionate about learning and applying new technologies in the development of projects in the field of engineering and pedagogy. I am a person committed to his job, where i seek to work in companies that promote research in order to continue developing professionally, learning new specialties, while contributing with my experience

JAMES MURILLO

EMBEDDED HARDWARE ENGINEER

EDUCATION

INDUSTRIAL UNIVERSITY OF SANTANDER(COLOMBIA)

Electronic engineer (2009 - 2018)

https://www.uis.edu.co

INVESTIGATION GROUP

Investigation Group ERA
(Electronics and applied robotics)

I had the oppotunity to enter the robotics group at the university developing my knowledge to a greater extent, since we allowed ourselves to develop robots and compete nationally and internationally in championships, having the convenience of working as a team and winnig different tournaments in the categories of : sumo, line follower and resolving maze.
https://n9.cl/0j189, (UDI 2012)

EXPERIENCIE

Freelancer
Project Developer
2012 – TODAY
Provided freelance services in electronic engineering, helping clients bring their product ideas and technical projects to life, from design to implementation.

Chiquimakers, Science Technology and innovation
Technology Teacher
https://chiquimakers.com
2016 – 2018
Steam education teacher, using technological tools, so that boys and girls learn in a gamified way concepts of design, engineering, mathematics, 3D printing, etc.

Conexalab
Lead Hardware Developer
https://www.conexalab.com
2018 – 2022
Development of didactic platforms for the learning of new information technologies in children and young people.

T.I.P
Specialized Technical Support
www.tipcolombia.com
2019 – 2020
Development in hardware and programming in needs raised by the oil & gas sector.

SEI Colombia
Hardware Consultant
https://seicolombia.com
2021 – 2021
Repair of electronic instrumentation for the oil & gas sector

WMSAS
Hardware Consultant
https://wmsas.co/site/
2021 – 2021
Consultant in electronic design of communication equipment

Freelance Robotics
Electronic Engineer
https://www.freelancerobotics.com.au
2023 – TODAY
Development of electronic systems involving PCB design, repair and calibration of electronic instrumentation, and firmware development for microcontrollers.

PROJECTS

- Graduation Project : Design of a functional model of a robot that favors the learning of basic concepts of programming in children.

  Resume: The project aims to engage children from 7 years of age, without any prior knowledge of programming. children must structure an algorithm, so that a physical robot that satisfies the levels set.
  Reference link : https://n9.cl/svjxd

- Collaboration, industrial design school : Device to quatify the strength and visual motor coordination in hand and finger rahabilitation therapies.

  Resume: Therapy centers do not have a device capable of evaluating the progress of people, who have suffered an upper limb work related accident. Therefore, a system was proposed that would allow quantifying the strength in the fingers and gamifying these therapies through games through an interface.
  Reference link : https://n9.cl/olux8

- Collaboration, conexalab : MAMUS, future knowledge community

  Resume: Through teamwork with the company conexalab, it was possible to develop a didactic platform with a pedagogical scope for boys and girls. The developed platform is called OCTABIO, an electronic card that allows the integration of a large number of sensors and actuators to develop a variety of projects.
  This educational system was created to generate social awareness at an early age, covering projects such as : energy generation using solar panels, irrigation system, prevention of noise pollution, fire alarm, intelligent mobility, etc. With this project, it is sought that thousands of young colombians who do not have easy access to information can get a kit and the pertinent accompaniment to transform the country and be leaders in development.
  Reference link : https://n9.cl/jw2dq

SOFT SKILLS



CERTIFICATIONS

- **Innovat contest 2018:** the event organized by the ecopetrol, had the challenge of developing a miniaturized communication module, it participated in the event with the “KUBIK” initiative, multipoint tracking system, managing to be among the finalists of the event.
- **Robotics contests 2011-2016:** in the course of my career i had the opportunity to compete in different categories, where the abilities and skills os each team in the fields of hardware and programming were put into play. having national recognition and winning an international contest in mexico city.
- **First Lego League Coach 2017:** As the lead technology instructor at Chiquimakers, I guided a group of children in the preparation and participation of the second First Lego League tournament in Colombia, Throughout the preparation, I enhanced their skills in robotics, programming, and teamwork, achieving a strong performance at the national competition.

TECHNICAL SKILLS

- Programming (C/C++ , Python) 75%
- 3D Design and Modeling(Fusion 360) 35%
- Hardware Design 90%
- PCB desgin (Altium, Eagle) 90%
- Soldering and Test Equipment Handling 95%
- Linux Systems Handling 45%

PERSONAL REFERENCES

- Andrea Nathalia Amaya Ruiz
  Industrial Designer
  Tel : +61 410768513
- Pablo Pieschacon
  Industrial Engineer
  Tel : +61 416893925

PROFESSIONAL REFERENCES

- Johan Suarez Largo
  Engineer Electronic
  Tel : +61 406280017
- Santiago Toloza
  Engineer Electronic
  Tel : +61 410752809

HOBBIES

Movies / Series · GYM · Running · Videogames · Futbol · Woodworking

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Australia

James Murillo

Embedded Hardware Engineer | Electronic Engineer

EXPERIENCE

Freelance Robotics, Australia - Electronic Engineer

MAY 2023 - PRESENT

At Freelance Robotics, I work as an Electronic Engineer, supporting the design, development, testing, troubleshooting, and repair of electronic systems and embedded hardware for different engineering projects.

My responsibilities include PCB design, electronic board assembly, component-level troubleshooting, repair of electronic modules, soldering and rework of SMD components, use of electronic test equipment, firmware development, and integration of embedded systems with sensors, actuators, communication modules, and software applications.

- **Smart caravan embedded system:** developed an embedded electronic system to control lights, refrigeration, battery monitoring, battery charging status, environmental variables, and other functionalities. I programmed the required modules and ensured that the generated data packets were correctly transmitted to the software application used to monitor and control the system variables.
- **Agricultural weed detection platform:** worked on the development of an electronic board for a weed detection system, integrating hardware components such as vision cameras, LiDAR, IMU, GPS, and other sensing modules. My role involved designing a six-layer PCB, supporting the hardware architecture, integrating electronic subsystems, and developing firmware to ensure the correct operation of the board within the complete detection platform.
- **Odour measurement system:** contributed to the development of an environmental monitoring system in collaboration with mechanical engineering colleagues. My role included PLC programming, integration of pressure sensors, MFCs, PPB measurement requirements, system control logic, and development of a custom graphical user interface using Visual Studio and C# for monitoring and controlling the system variables.

Conexalab, Colombia - Lead Hardware Developer / Project Lead

MARCH 2018 - OCTOBER 2022

At Conexalab, I worked as the lead engineer in the development of an educational robotics platform designed to help children and young people learn programming, electronics, and robotics through a low-cost and accessible learning kit.

My role involved defining the technical direction of the project, developing the hardware of the robotic platform, selecting electronic components, supporting firmware development, and contributing to the

design of the educational ecosystem around the product.

I also participated in the creation of learning content, project-based activities, and gamified resources to make technical concepts easier to understand for children and adolescents. The objective was to create a complete educational experience where students could learn theoretical concepts and apply them through practical robotics and programming challenges.

Freelancer, Colombia / Remote - Electronic Engineering Project Support

FEBRUARY 2012 - AUGUST 2022

During my studies in Electronic Engineering, I provided freelance technical support and guidance for academic and technical projects related to electronics, embedded systems, automation, and software development.

I assisted students and professionals from different engineering backgrounds, including Mechanical Engineering, Electronic Engineering, Mechatronics, and related technical fields. My role involved helping them understand project requirements, define technical solutions, select electronic components, develop basic circuits, support microcontroller programming, and troubleshoot hardware and software issues.

I also supported the development of prototypes and proof-of-concept systems, contributing to electronic design, firmware development, testing, and documentation. These projects involved practical work in circuit design, programming, sensors, actuators, and embedded systems.

Chiquimakers, Colombia - STEAM Technology Tutor

JUNE 2016 - FEBRUARY 2018

At Chiquimakers, I worked as a STEAM technology tutor for children and adolescents, teaching topics related to programming, basic electronics, circuit design, robotics, and mechanical prototype development.

The main purpose of this role was to guide students through project-based learning. For each topic, students learned the fundamental concepts and then applied them by developing a practical project, allowing them to connect theory with real implementation.

I was also involved in proposing class topics, designing new educational projects, preparing learning activities, and supporting students during the development and testing of their prototypes. As part of this experience, I also acted as a tutor and coach for students participating in the FIRST LEGO League robotics competition.

T.I.P, Colombia - Electronic Engineering Developer

SEPTEMBER 2018 - DECEMBER 2018 / SEPTEMBER 2019 - DECEMBER 2019

At T.I.P, I worked as part of the electronic engineering development team on projects related to the oil and gas sector, supporting technical solutions for pipeline inspection applications.

My role focused on the development of electronic hardware for a pipeline inspection tool designed to detect cracks or fissures and collect inspection data during the internal travel of a PIG system through oil pipelines. Based on previous design references, I contributed to the implementation of a more robust electronic platform capable of sensing defects and storing the acquired data in external memory for later analysis.

As an electronic engineering developer, I was involved in the design, implementation, and testing of electronic boards used for signal acquisition, data storage, and system integration. I also worked on power management considerations, as the system required approximately eight hours of autonomy during each inspection run while operating under space limitations that restricted the number of batteries that could be used.

SEI Colombia, Colombia - Electronic Board Repair Technician / Electronic Engineering Support

FEBRUARY 2021 - JUNE 2021

At SEI Colombia, I provided electronic repair and technical support services for control boards used in oilfield equipment and industrial machinery related to the oil and gas sector.

My role involved troubleshooting defective electronic boards, identifying damaged components, performing repairs, and verifying that the boards were operating correctly before being returned to service. I also supported testing procedures to ensure that the repaired boards met the required functional conditions for continued operation in the field.

This work required the use of electronic test equipment, component-level fault diagnosis, soldering, board inspection, and practical knowledge of electronic circuits used in industrial control systems.

WMSAS, Colombia - Electronic Hardware Consultant

SEPTEMBER 2021 - NOVEMBER 2021

At WMSAS, I provided electronic engineering consulting services for the development of an electronic tool related to smart metering systems.

My role focused on supporting the improvement of the system's energy efficiency, with the objective of extending battery life and allowing the device to operate in the field for long periods without frequent battery replacement.

I contributed to the analysis of the electronic design, power consumption requirements, and possible hardware improvements to optimise energy usage. This work involved reviewing the system architecture, identifying power consumption limitations, and proposing technical adjustments to improve the autonomy and reliability of the device during field operation.